THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 5: **Prime Time** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is divisible by 4? 1

(a) 3516 **(b)** 3518 **(c)** 3517 **(d)** 3519

2. Which pair is a pair of co-prime numbers? 1

(a) 6 and 9 **(b)** 12 and 18 **(c)** 10 and 25 **(d)** 8 and 15

3. Which of these is a pair of twin primes? 1

(a) 11 and 13 **(b)** 7 and 11 **(c)** 13 and 17 **(d)** 19 and 23

4. The number of prime numbers between 10 and 30 is: 1

(a) 7 **(b)** 8 **(c)** 6 **(d)** 5

5. The prime factorisation of 72 has how many prime factors (counting repeats)? 1

(a) 4 **(b)** 5 **(c)** 2 **(d)** 6

6. Using the Sieve of Eratosthenes, which is the first prime whose multiples we cross out after crossing multiples of 2? 1

(a) 1 **(b)** 5 **(c)** 3 **(d)** 4

7. The smallest prime number that is a factor of 105 is: 1

(a) 3 **(b)** 7 **(c)** 1 **(d)** 5

8. Which of the following is divisible by 8? 1

(a) 5130 **(b)** 5128 **(c)** 5131 **(d)** 5129

9. Which pair is a pair of co-prime numbers? 1

(a) 9 and 15 **(b)** 14 and 21 **(c)** 16 and 24 **(d)** 14 and 25

10. Which of these is a pair of twin primes? 1

(a) 17 and 19 **(b)** 31 and 37 **(c)** 19 and 23 **(d)** 23 and 29

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Test whether 3216 is divisible by 4 and by 8. 2

12. Write two prime numbers whose sum is 24. 2

13. Are 14 and 15 co-prime? Give the reason. 2

14. Test whether 7424 is divisible by 4 and by 8. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Use the divisibility test for 11 to check whether 2728 is divisible by 11. 3

16. Draw a factor tree (describe) for 96 and write its prime factorisation. 3

17. Find the first 4 common multiples of 4 and 6. 3

OR

Use the divisibility test for 11 to check whether 5412 is divisible by 11. 3

18. Draw a factor tree (describe) for 108 and write its prime factorisation. 3

OR

Find the first 3 common multiples of 6 and 8. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Find all prime numbers between 1 and 40 using the Sieve of Eratosthenes idea. State which primes you cross out multiples of. 5

OR

A number is divisible by both 3 and 4. Show that it is divisible by 12 when 3 and 4 are co-prime. Give an example with 60. 5

20. Find all prime numbers between 41 and 80 using the Sieve of Eratosthenes idea. State which primes you cross out multiples of. 5

OR

A number is divisible by both 4 and 5. Show that it is divisible by 20 when 4 and 5 are co-prime. Give an example with 140. 5

END OF QUESTION PAPER — Total: 40 Marks

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